

Gandharva Appagere

EXPERIMENTAL PHYSICIST • DETECTORS, DATA & AI

PhD candidate in experimental nuclear & hadron physics, Stockholm University

Stockholm
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PROFILE

Experimental physicist (PhD candidate, Stockholm University) working on a large accelerator experiment at GSI/FAIR. Detector work, particle identification, Monte Carlo simulation, statistical analysis, machine-learning classification, and automated analysis pipelines on HPC. Earlier industry experience in automotive radar algorithms and electromagnetic engineering.

EXPERIENCE

PhD Candidate (Doktorand) Aug 2021 - present
Stockholm University • *PANDA@HADES, GSI/FAIR* Stockholm, Sweden & Darmstadt, Germany
Experimental hadron physics on the HADES spectrometer at GSI and simulation studies for the CBM experiment at FAIR.

- $\Lambda\Lambda$ analysis in PANDA@HADES: reconstruction, PID, fits and systematics.
- Detector shifts (2022 proton and 2025 pion beam times); drift-chamber QA.
- Relative time-of-flight PID for start-detector timing issues.
- Multiclass ML classifiers (ROOT TMVA) for signal-background separation.
- Automated analysis pipeline on the GSI batch farm (Slurm / HPC), with LLMs.
- Event generator and detector-response folding for CBM ΞN -cusp studies.

EARLIER EXPERIENCE

Electric Propulsion Engineer, KTH Hyperloop team, Stockholm. High-energy propulsion unit for the SpaceX Hyperloop Pod Competition 2019. Sep 2018 - 2019

Associate Software Engineer, Robert Bosch Engineering and Business Solutions (RBEI), Bengaluru. Algorithms for automotive radar sensors. Aug 2017 - Jul 2018

Intern, Network Engineering Facility, AT&T, Bengaluru. VPN order-management processes in a network-engineering organisation. Jan-Jul 2017

Hybrid levitation for Hyperloop: engineering project, BMS College of Engineering, Bengaluru. Design of air-bearing levitation combined with linear-induction-motor propulsion for a Hyperloop concept. Aug 2016 - Jun 2017

Intern, Quality Assurance division, ISRO, Laboratory for Electro-Optics Systems (LEOS), Bengaluru. Circuit analysis of star, Earth and Sun sensors for the ASTROSAT and Chandrayaan-2 missions. Jul 2015

EDUCATION

PhD, Experimental Nuclear & Hadron Physics • *Stockholm University* 2021-2027 (expected)

Licentiate of Philosophy (Fil.lic.), Physics • *Stockholm University* 2021-2024
Thesis: "A- Λ hyperons: the road to estimate the production cross-section in proton-proton collisions"

MSc, Electromagnetics, Fusion and Space Engineering • *KTH Royal Institute of Technology* 2018-2020

BE, Electronics and Communication Engineering • *BMS College of Engineering* 2013-2017

CORE SKILLS

Data analysis, ML & AI Statistical inference • Uncertainty quantification • Kinematic (constrained) fitting • Machine-learning classification • AI-assisted development

Programming & software Python • C++ & ROOT • Git / GitLab

Simulation & modelling Monte Carlo event generation • Detector simulation • Finite-element electromagnetics

Detectors & instrumentation Tracking & timing detectors • Gamma spectroscopy • Beam-time operations & online data-quality monitoring • Radiation safety & radioactive-source handling • Automotive radar signal processing

Computing & HPC HPC batch computing • Containers & software distribution • Large datasets

SELECTED PUBLICATIONS & TALKS

"Investigating Double-Strange Systems at FAIR: $\Lambda\Lambda$ interactions and neutron star cores". **IN PRESS** 2026

"Hadron Physics Opportunities at FAIR". **PREPRINT** 2025

9 conference talks, including NSTAR 2026 (2026), 15th Nordic Meeting on Nuclear Physics (2026) and DPG Spring Meeting 2026 (2026). 2022-2026

LABORATORY, TEACHING & LEADERSHIP

Lab responsible, Nuclear physics student laboratory, Fysikum. 2021-2025

Chair, Fysikum PhD Council, Stockholm University. 2025-2026

Vice-Chair, Central Doctoral Student Council (CDR), Stockholm University. 2025

Vice-Chair, Natural Sciences Faculty Doctoral Student Council (NFR), Stockholm University. 2023-2024

Vice-Chair, Fysikum PhD Council, Stockholm University. 2023-2025

Doctoral-student representative, Fysikum Department Board and committees (Research Education, Technical Division, Computer, Work Environment). 2023-2026

Nuclear Physics (MSc) (FK7051) • Laboratory supervisor 2021-2025

Programmering, numeriska metoder och statistik för fysiker (FK4026) • Laboratory assistant 2021-2023

Detector Physics (MSc) (FK7056) • Laboratory supervisor 2024

Project in Experimental Modern Physics (FK5026) • Laboratory supervisor 2024-2025

Elektromagnetism och vågor (FK5019) • Lectures and tutorials on the concepts, with problem solving 2025

Vågrörelselära och optik (Waves and Optics) • Responsibility for assisting in exercise groups (tutorials and problem solving) and in labs (workout sessions on upcoming assignments) 2025

Bachelor's thesis in Medical Radiation Physics • Co-supervisor 2022

AWARDS & LANGUAGES

Awards & grants SAE E-Baja 2017: 3rd place in India, design review (2017); IEEE Darrel Chong Student Activity Award (2016); SAE Aero Design West 2015: 13th worldwide (team captain) (2015); 4 travel and research grants (2022-2026)

Languages English: Fluent • Swedish: Basic (A2)